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Warrant-Disciplined Agent Memory: Architecture, Instrumentation, and a Pre-Repair Baseline

Version 1 — FROZEN 2026-07-31. Written 2026-07-30 under M-memory-retrieval WS6. Intended audience: technical readers outside the futon stack (first: Rob). Sending gate: Joe.

Scope note added on freezing. The observation-channel figures in §5.5 (14 offered halves, 14% outcome completion, 71% empty) are correct for the corpus this draft could see: the author=ground-control slice of pattern-outcome receipts as of 2026-07-28. They are **not** representative of the system. An overnight run completed 2026-07-31 raised the corpus to 129 offered halves at **89% outcome completion** and 20 metric-bearing rows. Read §5.5 as a pre-repair baseline of a slice, and see holes/excursions/E-memory-whitepaper-v2-plan.md §1 for the measured before/after. Two further findings post-date this document and bear on it: recall loss is now diagnosed as structural (per-subject queries at 9–16 s against a 30 s total budget) and non-randomly biased toward analysis-heavy problems;

and bitemporal system-as-of is silently ignored by the store, which blocks the frozen benchmark of \$4.2 for existing dispatches. V1 is retained unedited apart from this note.

The chronological evidence ledger from which this draft was written — including every correction, retraction, and increment in the order it occurred — is preserved verbatim at docs/retrieval-evidence-ledger.md. This document is organised by argument; that one is organised by discovery date. Where the two disagree, the ledger records how we got here and this document records what we currently believe.

Provenance note: the mechanism described here has been developed in public repositories with dated commits (descriptions and working code, 2026-07-22 onward). This document consolidates that disclosure deliberately, serving as defensive publication. It is not legal advice; a US filing within the inventor grace period remains a separate decision, and Appendix A is written to keep that option cheap.

Abstract

An autonomous agent that stores what it learns does not thereby remember it: a memory that surfaces only when the agent happens to issue the right query is storage, not memory. We describe a deployed memory system in which retrieval runs from a task’s own text through independently reviewed edges to reusable patterns and their attached memories; in which every dispatch and every witnessed outcome writes a receipt; and in which those receipts — never self-report — update the retrieval coefficients, under an exploration-mass floor that guards against the loop collapsing onto its own early successes.

We state at the outset what this draft does and does not establish. The architecture is deployed and its learning loop has closed end to end under independent witness: a memory mined from failed sessions, recalled by pattern into a later session on the same problem, cited by the solver, and confirmed by a compiler. What is **not** yet established is that the combination outperforms its parts. The retrieval ablation that would settle it requires a frozen multi-query benchmark that does not yet exist, and the receipt-update comparison sits below its own preregistered calibration minimum at one metric-bearing observation.

What this draft contributes instead is an architecture, a set of falsifiable instruments, and an audited **pre-repair baseline** of the observation channel itself: across 14 frozen dispatch records, 71% of recalls surfaced nothing, outcome halves completed at 14%, and the field intended to record why a memory surfaced carries a vocabulary of exactly one string. A companion audit finds the field naming independent witnessing to be uncorrelated with the presence of witness records, and we exclude it from evaluation accordingly. Those numbers are reported as findings about our own instrument, not about retrieval, and they are what a forthcoming 489-problem run is being instrumented to fix. Two framing facts hold throughout: the system uses commodity agent models — the capability delta is the memory loop, not model scale — and it is operationally autonomous, with the human contribution confined to the strategic layer and the split auditable from the dispatch and receipt ledgers.

1. Problem and contributions

1.1 Storage is not memory

Agent memory systems fail in a characteristic way that has nothing to do with storage. Our own first deployment demonstrated it precisely: an agent deliberately recorded four typed memories from a proof session — correct, well-formed, full-text indexed — and they were operationally invisible, because nothing connected them to the moments where a later session would need them. A saved memory surfaced only if the agent independently thought to search for it, with the right words, at the right decision point. That is a diary, not a memory.

Three further failure modes constrain any fix.

Self-report contamination. If the system learns from the agent’s own claims of success, it learns to claim success. Every signal that updates retrieval must be independently witnessed — here, by a proof compiler.

Endogenous confirmation. A retrieval mechanism that learns which relations retrieve useful material will concentrate on relations that retrieve corroboration of what it already found: apparent certainty, hidden targets. Any learning retrieval loop needs an explicit guard, and the guard must be demonstrated against the failure it guards, not asserted.

Proposal/warrant confusion. Similarity — lexical or semantic — is cheap, and wrong often enough that letting it *justify* retrieval quietly replaces evidence with plausibility. Similarity may propose; only review may warrant.

The design question: what is the smallest mechanism that makes stored memories arrive unprompted at the decision points where they matter, gets better at this from real use, and resists the three failures above?

1.2 What this draft claims

Three primary claims are frozen for this draft. Each resolves to a row in the claim/evidence matrix (§4.5), and each is scoped to the evidence that currently exists rather than to the evidence we expect.

C1 — Warrant-disciplined conduction is implementable without sacrificing recall. Retrieval can be restricted to independently reviewed edges while similarity is retained as a proposal source, and the separation can be verified by mutation testing rather than asserted. *Status: supported, mechanism-level.*

C2 — Receipts make retrieval auditable, and the audit binds. An offered/outcome receipt channel makes retrieval behaviour inspectable after the fact, including its own failures; and a preregistered calibration gate over that channel refuses promotion when observations are insufficient. *Status: supported, including by the gate refusing itself.*

C3 — Structural retrieval features must pay functional rent, and our own instruments initially failed this test. Global topology statistics can be improved without improving retrieval; we demonstrate this on our own preregistered criterion, retract it, and replace it with a functional damage measure. *Status: supported as a negative methodological result.*

Two claims are explicitly **not** made in this draft, and are stated here so the reader does not have to infer their absence:

- That hybrid retrieval (content plus pattern) beats either route alone. The ablation exists as an offline instrument and has been run on two frozen cases; it has not been run on a benchmark (§4.2, §5.1).
- That receipt-updated operator coefficients improve ranking. The replay harness is validated and its live verdict is :below-calibration-minimum at $n = 1$ (§5.3).

1.3 Contributions

1. A deployed architecture combining typed bitemporal memory, pattern-mediated recall, warrant-disciplined conduction, a receipts observation channel, receipt-fed coefficient updates, and an exploration-mass floor (§2, Appendix A).
2. A set of falsifiability instruments that have each fired at least once against their authors: a confirmation-collapse battery, a calibration gate that refused promotion, a preregistered spectral prediction that failed and was diagnosed, and a functional-damage sweep that reframed what “damage” means (§5).
3. An audited pre-repair baseline of the observation channel (§5.5) and of the warrant fields (§5.6), reported as engineering findings about our own instruments, together with the instrumentation specification they imply (§8.2).

2. Deployed system and trust boundaries

2.1 Substrate

A memory is a **dialogue act** — assert, retract, challenge, agree — recorded as an append-only evidence body plus one typed hyperedge linking it to its subjects, the session and mission that produced it, the transcript spans it distills, and the *patterns* it instantiates. The act vocabulary is the IATC performative set.

The store is **bitemporal**: what is believed now and what was believed as-of T are both queryable. Correction is therefore an event with a timestamp rather than an overwrite, which is what makes the correction record in Appendix D a queryable object rather than a changelog.

2.2 Retrieval pipeline

Recall runs from the task’s own text, not from an agent’s decision to search:

```
problem text → term extraction → query ladder → text index
    → content matches —————
    → matched memories → their patterns → top pattern → attached memories
                                   |
                                   merge, dedupe, budget → candidates
```

Three properties of this path are load-bearing and each is discussed where it is measured:

- The **query ladder** (3-term conjunction, then pairs, then singles, first non-empty tier wins) exists because the text backend is conjunctive and composed queries died: measured on the live store, 1 term returned 5 hits, 3 returned 3, 7 returned 2, 12 returned 1, and 29 returned 0, while recall was composing up to 36 terms (§5.5).
- **Content matches merge ahead of pattern expansion.** Before 2026-07-30 they did not, and pattern arbitration could discard an exact content match — a failure traced end to end in §5.1.
- **Only reviewed edges conduct.** Text and embedding similarity may nominate; neither can warrant.

2.3 The four trust boundaries

The paper’s central discipline is that four distinct things are kept distinct. Conflating any two of them is the mechanism by which a memory system starts grading its own homework.

Boundary	What it establishes	What it does not establish
Review of an attachment	that an edge is well-formed and its ids resolve	that the memory is true or useful
Warrant for a memory	that evidence supports the memory’s content	that any agent used it
Attribution by the agent	that the solver <i>says</i> it used a memory	that the memory was load-bearing
Witness of the outcome	that a third party (a compiler) confirms the result	<i>which</i> memory produced the result

The fourth row is the one most easily elided. An independently witnessed proof does not independently witness the solver’s claim about which memory helped it. Throughout §5 we report attribution and witness separately and never let one stand in for the other.

A defect at this boundary, disclosed. The deployed :witness-status field is computed as a pure function of which scribe lane drafted the memory:

```
(if ({:solve-lane :arc-lane} lane) :independently-witnessed :self-asserted)
```

It therefore **cannot fail**: no property of the memory, the proof, or the receipt can make a solve-lane memory come out self-asserted.

An audit over the frozen mathematics projection (warrant-audit-20260730.edn, 62 attachments carrying the field) shows the consequence is not merely theoretical — the field is **uncorrelated with the evidence it names**:

	carries a witness record	carries none
:independently-witnessed (53)	1	52
:self-asserted (9)	1	8

Exactly two memories in the whole fixture carry a machine-checkable witness record — a list of Lean declarations with their proved results — and they fall **one on each side of the status split**. 98% of the memories the field calls independently witnessed carry no witness record at all, while the one memory whose body contains four verified declarations is labelled :self-asserted. A field that lands on both sides of the only two cases where evidence exists is not a weak signal; it is no signal.

(An earlier count over a different population — 59 solve- and arc-lane drafts, of which 28 carried :independently-witnessed with no witness record — is retained in the evidence ledger. The audit above supersedes it as the verifiable figure, being computed from a frozen, republishable artifact.)

Two things this is not. It is not a claim that the attestations are false — solve-lane memories distil proofs the compiler checked, and ground control re-elaborated #print axioms independently, so the underlying witnessing is real. Nor is it a review failure: edge review checks well-formedness and id resolution, and does so correctly. The defect is that the *status field* records a lane while the *evidence* for it lives outside the field’s derivation. **Consequently :witness-status is excluded as an independent evaluation signal everywhere in this paper**, and its repair is a draft-2 gate (§8.2).

2.4 Mechanism status table

Every mechanism referenced in this paper carries one of three statuses. Claims in §5 and §8 are made only at the level the status supports.

Mechanism	Status	Evidence
Typed bitemporal hypergraph store	deployed	§2.1
Pattern-mediated recall	deployed	§5.1
Query tier ladder	deployed (2026-07-30)	§5.5
Content matches merged ahead of patterns	deployed (50916c84, reviewed)	§5.1
Reviewed-edge warrant gate	deployed , mutation-tested	§5.4
Receipts (offered/outcome halves)	deployed	§5.5
:memory-use/surfacing-via attribution	deployed (2026-07-30, no data yet)	§5.5
Scalar per-memory Ψ (S6)	deployed	§5.3
Pattern-level Ψ -v2	implemented dark , below calibration	§5.3
k-step coupled propagation + floor	implemented dark , verified	§5.4
Beta-binomial per-pattern update (Rung 2)	implemented dark	Appendix A.2
Budgeted facet refinement (Rung 3)	implemented dark	Appendix A.3

Mechanism	Status	Evidence
Offline damage sweep (D_state)	implemented , run on 2 cases	\$5.2
Spectral admissibility criterion	retracted as a gate	\$5.2
Derived mid-tier technique clusters	proposed	\$5.2
Semantic proposal lane	chartered	\$8.2
Relation-type θ_r	proposed , degenerate at 1 edge type	\$5.3

3. Formal model

Only the notation needed to express implemented mechanisms and tested hypotheses is introduced. Where a mathematical object has no deployed counterpart, that is stated rather than glossed.

Retrieval state is $s_t = (x_t, \theta_t, F_t, B_t)$: activation over admissible memory nodes; coefficients over typed traversal operators; facet resolution; traversal budget. For relation-specific graph operators Δ_r , the pattern-conditioned operator is

$$\Delta_\theta = \Sigma_r \theta_r \Delta_r$$

and the coupled dynamics are

$$\begin{aligned} x_{t+1} &= \Phi(x_t, \Delta_{\{\theta_t\}}, q) && \text{(state update)} \\ \theta_{t+1} &= \Psi(\theta_t, x_{t+1}, q) && \text{(operator update)} \end{aligned}$$

— a diffusion whose Laplacian is itself updated by observation: “updating *which* Laplacian” (Rob, 2026-07-25), a second-order operator with location-dependent coefficients selected within a family. The degenerate regimes are named: **collapse** (all activation on one node — confirmation) and **dissipation** (flat activation — ranking nothing). The exploration-mass floor is the coefficient-space constraint that holds Δ_θ in the sustaining band between them.

3.1 Mathematical object → implementation status

This table is the honest core of the formal section. The deployed system is a considerably smaller object than the frame above.

Object	Deployed counterpart	Status
x_t propagation	one-step: text match → pattern endpoint → attached memories	deployed, k = 1
Δ_θ conductances	pattern attachments; editing a description edits a conductance	deployed
$\Sigma_r \theta_r \Delta_r$ (multi-relation)	recall traverses one relation (:memory/assert); the sum has one term	degenerate
θ_r per relation type	inactive: census sees a single edge type on the recall path	proposed
Ψ operator update	scalar per-memory ($1 + \alpha \cdot \text{used/offered}$), $\alpha = 0.5$	deployed (minimal shadow)
Ψ at pattern granularity	Ψ -v2 replay harness	dark, below calibration
Query source q	problem-file term extraction	deployed
Boundary conditions B_t	reviewed-edge warrant gate	deployed

Object	Deployed counterpart	Status
Forcing F_t	curriculum lane: coverage map + proposed targets	deployed
Exploration floor ε	lower-bounded simplex, $\varepsilon = 0.01$ live / 0.2 in battery	deployed in dark path
k-step coupled iteration	Rung 4 harness	dark, verified

The honest summary: **the deployed system instantiates the frame at $k = 1$ with a one-term operator sum and a scalar Ψ** . The frame is a research programme that the deployment shadows, and this draft claims results only for the shadow.

3.2 A structural correspondence, and where it stops

The 2026-07-30 runs exposed a failure whose shape was already named in our MetaCA work, and the correspondence is worth recording precisely because it also shows where the analogy stops.

Retrieval collapsed under **conjunctive composition** of query terms (§5.5). The remedy was not a better composite query but a *schedule* over the constituents — the tier ladder. Temporal braiding’s central observation has the same shape: *alternating* the elementary updates of two operators, rather than composing them into a single map, can sustain a field that either operator collapses on its own. In both settings the composed map is the dead one.

Where it does **not** hold: the ladder carries no state between tiers (no temporality); tiers do not reweight one another (no feedback); it short-circuits at the first non-empty tier rather than exposing a mixing fraction (a switch, not a dial). There is also a taxonomic caveat — the tiers are *subsets of one term set*, i.e. successive relaxations of a single operator, not two distinct operators alternating. Strictly this is decomposition-under-collapse, not braiding.

Taken as a research direction rather than a result, it points at a genuinely braided seam: alternate *structurally different* query operators (lexical, graph-endpoint, embedding); carry feedback so what tier n surfaces reweights tier $n+1$ (Ψ acting *within* a recall rather than between runs); and expose the mixing fraction as a precision/breadth dial. The first is half-present — the seam runs lexical and graph-endpoint stages — but they are composed in sequence, not braided, and their interaction has never been measured.

n is small and the effect is a plumbing repair, not evidence for a theory. The honest reading: a collapse-under-composition phenomenon appeared in a second, unrelated substrate, which is weak evidence that it is a property of conjunctive operator composition rather than of cellular automata.

4. Evaluation methodology

4.1 What exists and what does not

This draft evaluates against **frozen offline artifacts**, not against a benchmark. Stating the difference plainly is a precondition for reading §5 correctly.

Instrument	Artifact	Scope
Observation-channel audit	observation-channel-audit-20260730.edn	16 receipt halves
Warrant/provenance audit	warrant-audit-20260730.edn	62 attachments
Ψ -v2 dark replay	psi-v2-replay-results-20260728.edn	2 joined rows, 1 metric-bearing

Instrument	Artifact	Scope
Damage sweep (D_state)	damage-state-results-20260730.edn	2 frozen query cases
Connectivity meter	WS2 readings, stamps 20260727 / 20260730	whole reviewed graph
Collapse battery	WS1 battery file	1 synthetic corpus

The frozen relevance benchmark of Experiment 0 does not exist. Until it does, no claim in this paper is a benchmark result, and §5 is labelled accordingly throughout.

4.2 The benchmark this paper needs (specification)

Recorded here so draft 2 inherits a specification rather than a convention, and so the forthcoming run can be instrumented against it.

Construction. A chronological, transport-clean set of theorem-proving dispatches. For each query, preserve the corpus, retrieval graph, receipts, and retrieval implementation **as they existed at dispatch time**. Evaluating an old query against knowledge added only later is permitted only as an explicitly named counterfactual.

Labels. Each query/memory pair labelled *separately* for: relevance to the problem; whether it surfaced; whether the solver cited it; whether it was load-bearing in the resulting proof; whether the task outcome was independently witnessed. The four trust boundaries of §2.3 are why these are four labels and not one.

Exclusions. Timeouts, store outages, malformed responses, and unavailable transports are **missing or operational-failure observations, not retrieval misses**. §5.5 shows this is not a hypothetical concern.

Metrics (locked before arms are run): recall at a fixed candidate budget; reciprocal rank or nDCG; precision at budget; empty-result rate; retrieval latency.

Size. The existing $n \geq 20$ gate is an absolute minimum. Because the ablation is a *paired* design, the unit of evidence is the **discordant pair** — a query on which the arms disagree — not the query. A sign test needs at least 5 discordant pairs to reach $p < 0.05$ even when unanimous, and realistically 10–12 to survive a non-unanimous split. With discordance rate d , required $n \approx 12/d$; a 10–12 query pilot should measure d rather than assuming it.

Independence. The corpus is 10 pattern stars of sizes 30, 21, 14, 10, 4, 3, 3, 2, 2, 1, and concrete handles repeat heavily (one on 7 memories, another on 8, another on 6). Queries drawn from one theorem family are not independent: **effective n tracks the number of distinct families, not the number of queries**. The specification is therefore *n queries spanning ≥ 15 distinct theorem families, ≤ 3 per family*, analysed clustered by family.

4.3 Research questions

- **RQ1.** Does hybrid retrieval (content + pattern) improve relevant recall at a fixed budget over either route alone?
- **RQ2.** Which structural relations pay functional rent?
- **RQ3.** Do receipt-updated coefficients improve ranking on later observations?
- **RQ4.** Does the exploration floor prevent confirmation collapse?
- **RQ5.** How reliable is the observation channel itself?
- **RQ6.** Do the warrant and provenance fields record what they claim?

4.4 Preregistration and post-hoc discipline

Where a prediction was registered before a result, this is stated with the registering artifact. Two predictions in this paper failed and are reported as failures rather than repaired after the fact (§5.2, §5.3). One diagnosis of a failure was itself preregistered and re-tested (§5.2). The primary analysis does not change metrics or exclusions after seeing arm results.

4.5 Claim/evidence matrix

Claim	Current evidence	Required evidence	Status
Warrant discipline is implementable and verified	Mutation-tested review gate; content path reuses the same projection	Held-out confirmation at benchmark scale	Supported (C1)
Receipts make retrieval auditable, and the gate binds	Offered/outcome records; Ψ -v2 refused promotion at $n=1$; channel audit \$5.5	Receipt-completeness repair	Supported (C2)
Structural features must pay functional rent	Spectral criterion shown gameable <i>and</i> inverted; D_{state} sweep	$D_{functional}$ over the benchmark	Supported as negative result (C3)
The learning loop can close end to end	One witnessed chain (mine $\rightarrow$ recall $\rightarrow$ cite $\rightarrow$ compile)	Replication across problems	Single witnessed instance
Hybrid recall beats a single route	2 frozen cases showing complementarity at state level	Frozen multi-query ablation	Open (RQ1)
Receipt-updated operators improve ranking	Scalar deployed; dark replay at $n=1$	Chronological comparison, adequate observations	Below calibration (RQ3)
Exploration floor prevents collapse	Synthetic battery, both directions	Multi-seed and sensitivity replication	Mechanism evidence (RQ4)
The observation channel is reliable enough to evaluate on	Audited: 71% empty, 14% outcome completion, single-string attribution	Post-repair re-audit	Refuted for the pre-repair corpus (RQ5)
:witness-status records witnessing	Audited: uncorrelated with witness records, 1 vs 1 across 53/9	Derivation from evidence presence	Refuted; field excluded (RQ6)
Global topology predicts retrieval utility	Conflicting, projection-sensitive, criterion inverted	Functional predictive validation	Retracted as criterion

5. Results by research question

5.1 RQ1 — Hybrid retrieval: two cases, no benchmark

Result: not established. Complementarity demonstrated at state level on two frozen cases; generalisation untested.

The deployed pipeline had a defect that motivates the question sharply. `propose-patterns-by-query` searches evidence text, then converts matching rows into *pattern* proposals; `recall-by-endpoints` returns memories attached to the winning pattern. The pipeline was therefore

query $\rightarrow$ matching MEMORIES $\rightarrow$ their PATTERNS $\rightarrow$ top pattern $\rightarrow$ that pattern's memories

so a memory that actually matched was returned **only if its pattern also won**.

Durée run 25 dispatched on problem a92J05, a Rouché row. Eighty-seven seconds earlier, a memory named close-a92J05-by-transferring-the-unit-disk-zero-count — drafted from the immediately preceding attempt on *that same problem*, recording its exact residual goal — had been promoted, attached, and verified :attachment-status :reviewed. It did not surface. Five memories did, **all from a single pattern**; four were measure-theory memories the runner dismissed as irrelevant.

Replaying the dispatch’s own ladder shows retrieval was not the problem. The query was roots outside unit disk filter card; its first tier, "roots outside unit", returns 7 memories **including the a92J05 memory**, as do "roots outside", "outside unit", "unit disk" and "roots". The index found the right memory on essentially every tier. The matched memory sat on math/missing-dependency-protocol, the proposal resolved to tactic-algebra-interference, and the match was discarded in favour of five siblings of a different winner. **The system found the right answer and threw it away.**

The remedy — return content matches directly alongside pattern-mediated recall — was implemented and independently reviewed on 2026-07-30 (50916c84). Reviewed text matches now survive as :content-matches and merge ahead of endpoint recall; every surfaced memory carries :via :content-match | :pattern | :endpoint. Critically the content path does **not** widen the warrant set: it reuses the same reviewed-edge projection, and the negative test was verified non-vacuous by mutation — disabling the review gate turns it red, with an :attachment-status :unreviewed memory visibly leaking as a content match. This is the mutation evidence supporting **C1**.

The offline evidence for complementarity. The D_state sweep (damage_state_sweep.bb, fixture 20260730, write-once and byte-reproducible on rerun) ablates each arm on two frozen cases:

case	baseline	without content arm	without pattern arm
run 25	2 direct + 3 pattern	loses both direct matches, incl. the historically missed memory (Jaccard damage 0.571)	loses 3 pattern neighbours (0.600)
Lemniscate	architecture memory direct + 3 pattern neighbours	unchanged — the memory also returns via its pattern	loses all 3 neighbours (0.750)

The two arms are complementary in shape: direct matching protects exact nominations from arbitration; pattern expansion supplies related memories lexical search did not nominate. **Two cases are two cases.** They motivate RQ1; they do not answer it. Note also that the Lemniscate row is a case where the content arm was *redundant* — evidence that the complementarity is conditional, not universal.

5.2 RQ2 — Structural rent: a retracted criterion and a sparse damage profile

Result: the preregistered structural criterion is retracted. D_state measured on two cases; D_functional unmeasured.

This subsection reports a methodological failure of our own instrument, in three passes, because the sequence is the result.

Pass 1 — a preregistered prediction, run, and failed. The prediction: admitting the :subjects relation would raise λ_2 above the meter’s 0.1 floor with no new memories written. Measured against the frozen 20260730 export (method validated by reproducing the meter’s own reading, 0.0754 vs 0.0755):

graph	nodes	components	unnormalised λ_2	normalised λ_2
pattern + distills (baseline)	147	19	0.0754	0.0386
+ subjects as nodes	315	4	0.0836	0.0291
+ shared-subject projection	183	4	0.2002	0.0235

Admitting subjects as nodes does not clear the floor, and on the scale-robust normalised Laplacian it makes conduction *worse* (−24.5%). Concrete handles average 2.7 uses each and 116 of 161 are used once, so they attach pendant nodes rather than shortcuts. The prediction is **withdrawn rather than softened**.

Pass 2 — the repair exposed a defect in the instrument. Projecting the relation instead — joining two memories that share a handle — *does* clear the stated criterion, $0.0754 \rightarrow 0.2002$ against a 0.1 floor. But it clears it while normalised conductance *falls* by 39%. The reason is mechanical: the criterion is computed on the **unnormalised** Laplacian, where adding edges raises algebraic connectivity almost by construction, and a projection turns every k -memory handle into a k -clique. **A change that improves nothing about retrieval would have turned the meter green.**

Pass 3 — the metric is not merely gameable, it is inverted. Review by a second agent (E-memory-topology, Codex-3) made the decisive objection: a handle touching k memories is **one incidence relation, not $k(k-1)/2$ memory relations**, so the clique expansion was never the right operator. Recomputing with the degree-normalised hypergraph Laplacian (Zhou; nodes = memories, hyperedges = handles, no clique expansion):

operator	largest component	hyperedges in it	λ_2
patterns only	33 memories	1	1.0000
patterns + concrete subjects	126 memories	51	0.0689

The patterns-only figure is $\lambda_2 = 1.0$ because its largest component *is a single hyperedge* — one pattern containing 33 memories. A single blob attains maximal λ_2 by construction. Admitting topical handles raises reach from 33 to 126 memories, cuts components from 15 to 3, and **lowers λ_2 precisely because the structure stops being trivial**.

In this regime λ_2 is *anti-correlated* with useful structure. The criterion does not merely admit edge inflation: **it awards full marks to the most degenerate topology available**. The 2026-07-27 reading of $\lambda_2 = 0.99999$ was recorded as the one criterion that passed; it passed because the graph was a 6-node star. **Embodiment A.5 (spectral admissibility as a retrieval gate) is retracted**. Global spectral quantities may be reported descriptively; they gate nothing. This is the evidence supporting **C3**.

What survives as a positive result is **reach**, which is retrieval-relevant in a way spectra are not: the topical relation makes four times as many reviewed memories reachable from a seed. Whether that improves retrieval is a held-out question no spectral quantity settles. The governing rule adopted from E-memory-topology: *a structural change is good only when it improves held-out retrieval or witnessed outcomes at the same read budget, while preserving domain, temporal, review, provenance and witness invariants*.

The replacement measure. For an intervention A , functional damage is

$$D_{\text{functional}}(A) = \text{score}(G) - \text{score}(G \setminus A)$$

against a held-out retrieval score. The first sweep measures only the state-level precursor D_{state} : divergence in the ordered candidate set under exact frozen perturbation. Its profile is **sparse but occasionally large** — only 5 of 55 single-edge removals change the top-five state, and only 1 of 55 pattern-role removals does. The exception is instructive: removing the attachment of finish-an-outside-root-count-from-an-inside-count-and-no-boundary-roots to tactic-algebra-interference changes three of five candidates (Jaccard damage 0.750); removing the whole edge changes four of five (0.889).

This yields a more precise causal object than “a memory node”: the relevant unit is a **nomination attachment**, whose removal changes which whole neighbourhood consumes a bounded candidate budget. It is a discontinuity in proposal selection, not diffusive propagation. In the Lemniscate case one content-matched attachment gates an entire cluster — removing its whole edge takes the candidate set from four ids to empty. The neighbourhood is internally useful by the historical receipt, but operational access to it is **brittle**.

Whether any of this changes proof outcomes is D_functional and is unmeasured. No structural score should be fitted to two cases.

5.3 RQ3 — Receipt updates: the calibration gate binds

Result: :below-calibration-minimum at n = 1. The mechanism is validated; the claim is refused.

Ψ -v2 moves the coefficient from the node (memory) to the edge structure recall traverses. Credit assignment is the one non-obvious decision: receipts do not record which pattern surfaced each memory, so attribution is *reconstructed* deterministically — restrict a memory’s reviewed attachments to those whose pattern terms intersect the recorded query, fall back to all attachments if empty, split one unit of credit uniformly, and report the mode (:matched / :fallback-all / :unattributable) per row. Fractional credit is fine; invented precision is not, so there is no similarity weighting.

The update is

$$\begin{aligned} \theta_p &= 1 + \alpha \cdot (\text{used}_p / \text{offered}_p) && \text{if } \text{offered}_p \geq \text{n-min-coeff} \\ \theta_p &= 1 && \text{otherwise (per-coefficient abstention)} \end{aligned}$$

with $\alpha = 0.5$, bounded, cold-start neutral, and passed through the lower-bounded simplex with explicit floor ε .

Two gates, deliberately different. Harness activation is $\text{n-min-coeff} = 5.0$ fractional offered credits. Live promotion is the Phase 6 standard: **$n \geq 20$ independently witnessed outcomes per coefficient**, plus interface coordination at a cohort boundary.

The live result. Three arms (no- Ψ / deployed scalar / pattern-level) ran leave-one-out over the frozen receipt corpus. Two rows joined; one has non-empty used-ids. All three arms score MRR 1.0 and hit@1 1.0 — which says only that all arms retained an already-correct top item in one row. The other joined row is :surfaced-not-usable and is excluded from MRR as preregistered. No scored row was self-fitted. Twenty pattern coefficients are reported; **none reaches the activation minimum of 5.0**, the largest observed counts being 2.0 offered / 2.0 used.

The census found 51 :pattern-attachment edges and no second edge type, so relation-level θ_r is correctly marked :inactive-degenerate — the sum $\sum_r \theta_r \Delta_r$ has one term and the exploration floor constrains a singleton.

The ranking contract is nevertheless proven correct, on a synthetic fixture whose answer is computable by hand: six offered and six used credits to a planted good-route coefficient yield $\theta = 1.5$ and deterministically flip the ranking from [b1, g1] to [g1, b1], asserted in the harness rather than eyeballed.

A system that publishes its own “not enough evidence yet” verdict alongside a proven-correct mechanism is the evidentiary posture this document maintains throughout. **This is the evidence supporting C2:** the gate is real because it refused its author.

5.4 RQ4 — The exploration floor: mechanism evidence, single configuration

Result: demonstrated in both directions on one synthetic corpus. Multi-seed replication outstanding.

On a synthetic corpus with a planted target and a decoy relation seeded with early accidental corroboration — expectations preregistered in the battery file before the run — the floor-off ablation collapses as the theory predicts: θ concentrates entirely on the decoy ({repairs

1.0, requires 0.0}), hiding the planted target. The identical system with $\varepsilon = 0.2$ recovers the planted target at step 2.

Critically, *both* control arms — fixed endpoint order and one-step typed ranking — rank the decoy first on this corpus. The recovery is produced by the iterated coupled dynamics under the floor, not inherited from any single-step method. Independently witnessed challenge memories remained reachable at every step of every run; all runs deterministic and fully audited, with $k = 1$ identity against the one-step ranker holding exactly.

This is a guard that first demonstrates the failure it guards against, which is the correct standard. It is also **one corpus, one seed, one decoy strength, one floor value**. The claim is narrow by construction: *under the constructed dynamics, the exploration floor prevents or reverses collapse*. Sensitivity across seeds, decoy strengths, floor values and traversal budgets is a draft-2 requirement (§8.2).

A related methodological yield from the same workstream: a preregistered check that “diffusion time ranks inversely with λ_2 ” scored Spearman $\rho = 0.0$ and was retained as a failure. The diagnosis — three of four configurations sat past the explicit-Euler stability boundary $\varepsilon < 2/\lambda_{\max}$, so those runs measured concentration (power iteration), not diffusion — was itself preregistered and re-tested at $\varepsilon = 0.1$, giving $\rho = -0.8$, confirmed. **Step size is part of the operator**: any deployment of iterated retrieval dynamics must report ε against the spectral radius or declare the concentration regime deliberately.

5.5 RQ5 — The observation channel: an audited pre-repair baseline

Result: the channel is the binding constraint on every other experiment. Reported here as an engineering finding about our own instrument.

This is the one research question this draft answers at full strength, and the answer is unflattering. Audit artifact: observation-channel-audit-20260730.edn, produced by observation_channel_audit.py over the frozen 20260728 export; read-only and deterministic. It reproduces the WS3 join numbers independently.

Stage	Count	Rate
Offered halves (dispatches)	14	—
... of which recall returned :ok	4	29%
... of which :recall-empty	10	71%
Outcome halves	2	14% of offered
Joined offered→outcome rows	2	14%
Metric-bearing rows (non-empty used-ids)	1	7%
Total memories surfaced across all 14 dispatches	8	0.6/dispatch

Three findings.

(i) Missing observations are not negative observations. Of the 10 empty recalls, 2 carry :recall-reason :timeout in this export; a wider read of the lane records 21 :timeout and 13 :store-unavailable. The dispatch path collapsed :timeout, :store-unavailable and a genuine empty result into a single runner-visible signal — “no dispatch-time memories were supplied” — so the runner’s report cannot distinguish them. A substantial share of this lane’s recorded “recall surfaced nothing” observations are **infrastructure failures rather than retrieval evidence**.

This is not hypothetical, and it caught the present author. An earlier draft of §5.1 explained a run that surfaced zero memories — despite six reviewed memories carrying its exact subject handle — as a fragmentation effect. The persisted offered-half record shows :recall-reason :timeout. Recall never completed; nothing was arbitrated, fragmented or discarded. **An infrastructure failure was diagnosed as an architectural one.** The arbitration finding of §5.1 is unaffected — that dispatch recorded :recall-status :ok and returned five memories — but the two runs are adjacent on the same problem and produced the same runner-facing sentence for entirely different reasons.

(ii) The attribution field is populated and uninformative. All four rows that surfaced anything carry `:inclusion-reasons`. Across every entry in the corpus the **vocabulary has size one**: "reviewed attachment surfaced by terrain-conditioned dispatch recall". The field is present, well-formed, and cannot distinguish which pattern surfaced which memory — which is precisely why Ψ -v2 must reconstruct attribution lexically (§5.3). (*This corrects the Ψ -v2 design note, which records the field as empty in live rows; it is not empty, it is constant. The practical consequence is the same, the repair is not.*) The replacement — `:memory-use/surfacing-via`, recording `:content-match | :pattern | :endpoint` per memory — is **deployed as of 2026-07-30** and carries no data in this export because the export predates it.

(iii) The corpus was starved, and the causes are known and partly fixed. Recall was composing conjunctive queries of up to 36 terms against a backend whose hit count falls off a cliff with term count (1 term $\rightarrow$ 5 hits, 3 $\rightarrow$ 3, 7 $\rightarrow$ 2, 12 $\rightarrow$ 1, 29 $\rightarrow$ 0). Measured across five live rows using real term-extraction output, the composed 3-term query returned **zero** memories for all five, while the decomposed ladder surfaced a memory for **three**. Two rows correctly returned nothing at any tier. Two further defects from the same record: query terms are partly scraped from the dispatch packet's prose, so operator-written preamble dilutes the query; and the ladder's tier-selection predicate tests only `(:candidates p)`, ignoring `:content-matches`, so a tier finding a content match but no pattern candidate is skipped (latent — every tier with content matches on the traced row also had candidates).

A separate indexing finding: "a92J05" and "rouche" each return zero memories at the deployed limit, although a memory is *named* `close-a92J05-...` and eight carry the subject `rouche-root-count-transfer`. The surface is an application-controlled SQLite FTS5 sidecar (`futonlbi.text-index, tokenize='unicode61'`), where hyphens are ordinary separators. The causes are **bounded-result starvation** — the targets rank at 64 and 30 under `limit=100` and fall outside smaller deployed limits because coordination records outrank memories — and **body-only indexing**: only `:evidence/body` is indexed, the `id` column is UNINDEXED, and top-level `:subjects` are not indexed at all. The concrete handles carrying this corpus's topical structure are **absent from the index**. This is a retrieval-layer fact and must not be confused with the graph-layer claim retracted in §5.2: indexing subjects would make them *findable*, a different proposition from the refuted claim that relating them improves conductance. It has not been done.

Reading. The 20260728 export is a **pre-repair baseline**. The ladder, the content-match merge, and `surfacing-via` all landed on 2026-07-30, after it. Draft 2's first duty is to re-run this audit post-repair and report the delta.

5.6 RQ6 — Warrant fields: the status does not track the evidence

Result: refuted. `:witness-status` is excluded as an evaluation signal.

The full cross-tabulation and its consequences are given at §2.3, where the defect belongs architecturally. The evaluation-facing summary: across 62 attachments carrying the field, the two memories with actual machine-checkable witness records fall one on each side of the 53/9 status split, and 52 of 53 memories labelled `:independently-witnessed` carry no witness record. Artifact: `warrant-audit-20260730.edn`, regenerable with `warrant_audit.py`.

Two methodological notes, since this audit corrected itself twice in the course of being run and the corrections are the reason to trust it. A first pass reported 53 of 53 — a clean 100% — because the record-slicing walked balanced braces and never reached the memory body where witness records actually live. A second pass with a two-level walk reported the same figure for the same reason. Only a hand check of the one memory known to carry `:witnesses` revealed that the attachment and the body sit 1693 characters apart *at equal brace depth* in this serialization, so no brace walk of any depth would pair them. The audit now partitions on marker position instead, and its output reconciles exactly with the raw count of `:witnesses` occurrences in the file — two, which is the cross-check that a clean 100% would have failed.

The relevant lesson for the benchmark of §4.2 is that **a suspiciously clean number is a reason to check the instrument**, and that the check must be against a quantity derivable

independently of the instrument — here, a raw string count.

5.7 The loop has closed once, under witness

Reported separately because it is a single instance and should not be read as a rate.

The learning loop closed end to end on 2026-07-25/26: a memory mined from failed sessions — a lemma absent from the library — recalled by pattern into a revisit of the same problem, cited by the runner as redirecting its effort, producing a proven lemma and a sorry-count drop, with both receipt halves written and the outcome witnessed by the compiler rather than self-reported.

A second episode demonstrates **cross-model transfer**: two sessions by a codex runner on a construction target, dispatched with recall over a corpus mined from *zai* sessions. In both, all three surfaced memories were cited as used with specific behavioural effects; in the second, the target file reached zero remaining proof obligations, verified independently at the axiom level (no *sorryAx* transitively; standard axioms only). The decisive memory — a process rule recording a previously-hit API absence — carried a two-branch decision structure (“prove the local inequality, or declare the dependency frontier”), and the two sessions exercised *both* branches: the first declared and priced the frontier, the second proved the local result by the elementary route, making the frontier unnecessary. Portability is demonstrated at the level of decision structure, not merely recalled facts.

Both episodes are attribution-plus-witness: the compiler witnesses the *outcome*; the runner attributes the *use*. Per §2.3 these are different claims, and no instrument in this draft independently establishes that the cited memory caused the outcome. That is what *D_functional* over the benchmark is for.

6. Failure analysis and threats to validity

Sample size dominates everything. One metric-bearing row, two frozen damage cases, one witnessed loop closure, one battery configuration. No result in §5 except RQ5 rests on a sample that would survive an ordinary power calculation. Where an effect is reported, it is reported as a mechanism demonstration or a motivating case.

Attribution is not causation. The solver’s citation of a memory is self-report about *use*, made trustworthy at the outcome layer only because a compiler witnesses the *result*. A runner could cite a memory that did no work in a proof that succeeded for other reasons. Nothing in this draft excludes that.

:witness-status is structurally uninformative (§2.3, §5.6) and is excluded as an evaluation signal. Any future analysis that reads it before its repair inherits a field that cannot fail — and which, on audit, does not track the evidence it names. Note that this threat propagates backwards: any *earlier* analysis in the ledger that used the field is suspect, and the ledger is not retroactively corrected.

Theorem-family dependence. The corpus concentrates: 39% of memories sit behind one hub, and that hub — missing-dependency-protocol — is a *process* category, not a mathematical area. Results drawn from this corpus may reflect the shape of one curriculum rather than a property of the method.

Infrastructure missingness is non-random. Timeouts and store unavailability are more likely under load, which correlates with dispatch batch size, which correlates with which problems were being run. Treating these as missing-at-random would be unjustified.

Corpus quality bounds what recall can deliver. 22 of 90 memories (24%) carry a `:hook` identical to the memory name; the `:level` field has 40 distinct values over 91 drafts, 28 of them singletons. A type dimension whose vocabulary grows linearly with its instances is not typing anything. The hook figure was reached independently twice, over disjoint samples, which is why it is stated as a corpus property rather than a sampling artifact.

Dark mechanisms are not deployed mechanisms. Rungs 2–4 and Ψ -v2 are verified against their own contracts, not against production traffic. A contract-correct mechanism can still be the wrong mechanism.

Author $\neq$ reviewer is upheld but not blinded. The decisive objection of §5.2 came from a second agent, and the §5.5 correction from the implementing agent. Both are real independence; neither is blinded review.

7. Related work

The combination claimed here is novel; the ingredients are not, and positioning them accurately is part of the claim.

Spreading activation and cognitive architectures. Retrieval as propagation over an associative network descends from Collins and Loftus’s spreading-activation theory of semantic processing. The closest functional relative is ACT-R’s declarative memory, where retrieval probability depends on a base-level activation that rises with use and decays with time, plus associative strengths from context. Our x_t propagation and use-driven Ψ occupy the same conceptual position. The differences are architectural rather than mathematical: ACT-R’s activation is not gated by an independent review of the edges it traverses, and its strengthening signal comes from the model’s own retrieval history rather than from an external witness.

Relevance feedback and counterfactual learning to rank. Rocchio-style relevance feedback established that retrieval should be updated by evidence of use. The modern line — unbiased and counterfactual learning-to-rank from implicit feedback — confronts exactly our self-report problem in a different guise: click logs are biased by position and by what the ranker chose to show. That literature’s remedy is propensity weighting over a model of the presentation bias. Ours is different in kind: rather than de-biasing a self-generated signal, we require the update signal to be **independently witnessed** by a compiler, and treat the agent’s own attribution as a separate, weaker label (§2.3). The two approaches are complementary, and the propensity literature is the right source for what our receipts channel should record about *what was shown*, which is the reason surfacing-via matters.

Learned sparse and dense retrieval. DPR, ColBERT, and SPLADE learn the representation and let similarity do the work. Our position is the opposite on one axis: similarity may propose but never warrant (§2.3). This is a deliberate capability sacrifice — we forgo learned recall over unreviewed material — bought in exchange for an auditable warrant chain. Whether the trade is worth it is an empirical question this draft does not settle; the semantic proposal lane (chartered) is designed to test it by adding embedding proposals strictly under the review gate.

Graph-structured retrieval. Random-walk and personalised-PageRank retrieval over entity graphs, and more recently graph-structured RAG with community summarisation, share our premise that structure carries retrieval signal. Our §5.2 result is a caution addressed to that literature: we adopted a global spectral admissibility criterion, preregistered it, and found it not merely gameable by edge inflation but **anti-correlated** with useful structure in the sparse regime, awarding its maximum to a single-hyperedge blob. Structural quality metrics in graph retrieval need functional validation, not just topological plausibility.

Hypergraph spectral theory. The correction in §5.2 pass 3 turns on Zhou-style degree-normalised hypergraph Laplacians: a hyperedge over k nodes is one incidence relation, and clique expansion silently converts it into $k(k-1)/2$ relations, inflating connectivity by construction. The broader framing — retrieval as a diffusion whose operator coefficients are themselves updated — is Sturm-Liouville in spirit, and normalised spectral graph theory supplies the scale-robust quantities we should have used from the start.

Agent memory systems. MemGPT’s paging between context tiers, the memory-stream-plus-reflection architecture of generative agents, and skill-library approaches such as Voyager all address the same practical problem. Retrieval in these systems is typically scored

by a blend of recency, importance and embedding relevance, where importance is model-assigned. That is precisely the self-report channel §1.1 rules out. Our contribution relative to this line is the warrant discipline and the witnessed-outcome-only update, plus the receipts channel that makes both auditable.

Bandits and exploration floors. The exploration-mass floor is ϵ -greedy's guarantee applied in coefficient space rather than action space, and the minimum-exploration-probability construction in adversarial bandit algorithms is its nearest formal relative. What is unusual here is not the mechanism but the acceptance standard: the floor is required to first *demonstrate* the collapse it prevents (§5.4).

Provenance. The bitemporal store and receipt chain are a provenance graph in the W3C PROV sense, specialised so that the provenance is consumed by retrieval rather than merely recorded for audit.

Our own lineage. The MetaCA work supplies the coupled express-then-evolve structure of §3 and the braiding correspondence of §3.2, and the honest report is that the correspondence is structural rather than instantiated.

A full bibliography with verified citations is a draft-2 deliverable; this section names lineages and positions the contribution, and should not be read as a citation-complete survey.

8. Conclusion and research programme

8.1 What is established

The architecture is deployed and its parts hold under adversarial inspection. Warrant discipline is implemented and mutation-tested (C1). The receipts channel makes retrieval auditable, and its calibration gate demonstrated that it binds by refusing its author's own promotion at $n = 1$ (C2). The requirement that structural features pay functional rent is supported as a negative methodological result: our own preregistered spectral criterion was shown to be inverted in the sparse regime and is retracted (C3). The learning loop has closed end to end once under independent witness, and memory has transferred across runner models at the level of decision structure.

What is **not** established is that the combination outperforms its parts. That is not a hedge; it is the state of the evidence, and every instrument needed to settle it now exists offline.

8.2 Future work for draft 2

Draft 2 is gated on data that a forthcoming run of the 489-problem APM queue is expected to supply. This subsection is written to double as the **instrumentation specification** for that run: what it must capture, and what is unrecoverable if it does not.

8.2.1 Capture requirements (unrecoverable if missed) These cannot be reconstructed from a finished dispatch. Every problem that runs without them is a permanently degraded sample.

1. **Per-memory surfacing attribution.** `:memory-use/surfacing-via` is **deployed** (`dispatch_with_recall.clj:877`, `memory_recall.clj:363`) and supersedes the constant-vocabulary `:inclusion-reasons` field (§5.5(ii)). Draft 2 must verify it is populated in live rows before the run scales — a deployed writer is not the same as a populated field. Without it, Ψ -v2 credit assignment stays lexically reconstructed and every coefficient inherits attribution noise.
2. **Outcome-half completion.** Currently **2 of 14 (14%)**. This is the single highest-leverage repair in the system: it multiplies the yield of every dispatch, and an uncompleted half wastes a full solve. Target $\geq 90\%$, with the completion rate itself reported per cohort.
3. **Dispatch-time corpus snapshots.** Experiment 0 requires transport-clean chronological state (§4.2). The corpus grows *during* the run, so if the graph and index state

are not stamped as problems go by, dispatch-time state cannot be reconstructed afterwards and every query becomes contaminated by later knowledge.

4. **:recall-reason on every empty result.** Already partially present. Missing-vs-negative must be separable per row (§5.5(i)), or the benchmark’s exclusion rule cannot be applied.

8.2.2 Experiments unblocked by data, with their gates

Experiment	Gate	Projection at 489 dispatches
E1 Hybrid ablation (RQ1)	benchmark exists; ≥ 15 theorem families	Reachable — labelling, not accrual, is the cost
E2 D_functional (RQ2)	benchmark + relevance labels	Reachable; harness exists
E3 Receipt replay (RQ3)	$n\text{-min-coeff} = 5.0$ dark; $n \geq 20$ witnessed per coefficient live	See below
E4 Floor sensitivity (RQ4)	none — synthetic	Runnable now
E5 Channel re-audit (RQ5)	post-repair receipts	Runnable as soon as repairs land
E6 Warrant audit	:witness-status repair	Blocked on the repair

E3 arithmetic, at the audited rates. At the current 14% outcome-half completion, 489 dispatches yield ≈ 70 outcome halves and ≈ 35 metric-bearing rows. Credit distributes roughly as the pattern-star sizes (30/21/14/10/... over 90 memories), so only the **top one or two** coefficients would clear the $n \geq 20$ promotion minimum. At a repaired 90% completion the same 489 dispatches yield ≈ 440 outcome halves and the **top five or six** coefficients clear. **The completion repair is worth roughly five times as much as the dispatch volume it multiplies.**

Note that partial promotion is a legitimate result, not a fudge: the harness’s per-coefficient abstention ($\theta_p = 1$ below minimum) is designed for exactly this, so “top k coefficients promoted, tail abstained” is a reportable Threshold-B outcome.

8.2.3 Repairs draft 2 should report as done or explicitly deferred

- :witness-status derived from the presence and resolution of a witness record, so that a solve-lane memory lacking one is **reported rather than promoted** (§2.3).
- Text index: index :subjects and reconsider the UNINDEXED id column; raise or stratify the result limit so memories are not outranked out of the budget by coordination records (§5.5(iii)).
- Ladder tier-selection predicate to consider :content-matches, not only :candidates (§5.5(iii)).
- Term extraction to stop scraping dispatch-packet prose (§5.5(iii)).
- Connectivity meter to report normalised alongside unnormalised λ_2 , or be removed from the gate path entirely (§5.2).

8.2.4 Open research questions, unchanged by more data

- Whether the derived mid-tier technique clusters improve retrieval at fixed budget, tested as an intervention rather than assumed from cluster structure (§5.2).
- Whether a genuinely *braided* retrieval seam — structurally different operators alternating with feedback and a continuous mixing fraction — outperforms the current sequential composition (§3.2).
- Whether the warrant discipline’s capability sacrifice is worth its auditability, testable once the semantic proposal lane runs under the review gate (§7).

8.3 Closing

The honest summary of draft 1 is that we have built a memory system whose architecture is sound, whose instruments are sharp enough to have cut their authors six times over (Appendix C), and whose central effectiveness question is still open because the observation channel that would answer it was itself the thing most in need of repair. Reporting that in the body, rather than discovering it in review, is the point of the exercise.

Appendix A. Claims-grade specification

This appendix states the mechanism at the precision a patent filing would require. It is the defensive-publication payload; the body above is the document.

Independent claim — the combination. A memory system for autonomous agents comprising:

- **(a) A typed, bitemporal hypergraph store.** A memory is a dialogue-act record: one append-only evidence entry (the body, full-text indexed) plus one typed hyper-edge whose endpoints link the entry to its subjects, the session and mission that produced it, the transcript spans it distills, and the *patterns* it instantiates. Act vocabulary = IATC performatives (assert/retract/challenge/agree/define/...); bitemporal (valid-time/system-time) so belief-as-of-T, correction lag, and supersession are queryable objects, not file overwrites.
- **(b) Pattern-conditioned recall.** Retrieval is propagation, not lookup: query-derived seed terms $\rightarrow$ full-text match $\rightarrow$ typed edge to a pattern endpoint $\rightarrow$ the memories attached to that pattern. The pattern is the reusable retrieval handle; the memory stays a concrete episode.
- **(c) A warrant discipline (boundary conditions).** Only independently REVIEWED edges conduct retrieval. Lexical bridges (pattern descriptions) and semantic similarity (embeddings) may *propose* candidates; they never *warrant* them. Editing a pattern's description edits a conductance, not a warrant.
- **(d) A receipts observation channel.** Every dispatch writes an *offered* half (surfaced ids, query, term sources, recall-system version); every independently witnessed outcome writes an *outcome* half (used ids from the agent's own citation, result, witness refs). Reasoned non-use is a first-class outcome. The channel is never-blocking: recall failure cannot block dispatch.
- **(e) A receipt-consuming operator update (Ψ).** Retrieval coefficients are reweighted from the receipts channel by a bounded, cold-start- neutral, audited update rule — retrieval whose *operator* learns from witnessed use, not from self-report.
- **(f) An exploration-mass floor.** The coefficient vector retains a minimum mass $\varepsilon > 0$ on every admitted relation type at every step — an explicit, reported parameter guarding against endogenous confirmation collapse.

The claim is the *combination*: (a)–(b) alone is a graph memory; (c) alone is review policy; (d)–(e) without (c) learns from contaminated signal; (e) without (f) amplifies confirmation. Together they close a loop in which retrieval behaviour is updated only by independently witnessed outcomes, under warrants, with collapse guarded.

Dependent embodiments (status dated; see §2.4):

1. *Scalar per-memory Ψ* : multiplier $(1 + \alpha \cdot \text{used/offered})$, $\alpha = 0.5$, cold-start neutral, audited in the receipt. **Deployed** (S6).
2. *One-outcome per-pattern update*: θ multiplied by the posterior- probability ratio of a beta-binomial useful-progress model refit on exactly one new independently witnessed transition; promotion gated on a calibration minimum ($n \geq 20$). **Implemented dark** (Rung 2).

3. *Budgeted facet refinement*: next facet expansion chosen by expected information gain per cost, with child eligibility gated on witnessed parent-edge warrants. **Implemented dark** (Rung 3).
4. *k-step coupled propagation*: iterate x/θ updates under budget with per-step audit (entropies, path diversity, challenge reachability, explicit termination class) and the floor of (f); acceptance built on a confirmation-collapse battery — floor-off ablation must collapse, floor-on must recover a planted target. **Implemented dark, verified 2026-07-27** (Rung 4 / WS1; §5.4).
5. *Spectral admissibility diagnostics*: algebraic connectivity λ_2 and spectral gap as a threshold below which retrieval dynamics cannot beat direct lookup. **RETRACTED as a gate, 2026-07-30** (§5.2): the criterion is anti-correlated with useful structure in the sparse regime, awarding its maximum to a single-hyperedge component. Spectral quantities may be reported descriptively and gate nothing.
6. *Pattern-level receipt conductances (Ψ -v2)*: fractional credit assignment from receipts through reviewed attachments; per-coefficient abstention below an observation minimum; promotion gated at $n \geq 20$ per coefficient. **Implemented dark, verified 2026-07-28**; live verdict :below-calibration-minimum at $n = 1$ (§5.3).
7. *Semantic proposal lane*: embedding nearness as an additional proposal source strictly under (c). **Chartered** (WS4 / S5).
8. *Curriculum forcing*: coverage maps plus memory-proposed construction targets select what the system experiences next (the forcing term F_t). **Deployed** (cohort-2 curriculum lane).
9. *Bitemporal learning meters*: correction lag, belief-as-of-T, time-to-fill as queryable observables of the learning process itself. **Substrate deployed; meters partial**.

Appendix B. Artifacts and reproduction

Artifact	Path	Determinism
Receipts export (frozen)	holes/labs/M-memory-retrieval/receipts-export-20260728.edn	bounded read-only capture
Channel audit	holes/labs/M-memory-retrieval/observation-channel-audit-20260730.edn	deterministic; regenerate with observation_channel_audit.py
Warrant audit	holes/labs/M-memory-retrieval/warrant-audit-20260730.edn	deterministic; regenerate with warrant_audit.py; cross-checks against raw :witnesses count
Ψ -v2 replay	psi_v2_replay.bb, psi-v2-replay-results-20260728.edn	deterministic; ties broken by memory id
Damage sweep	damage_state_sweep.bb, damage-state-fixture-20260730.edn, damage-state-results-20260730.edn	write-once; byte-reproducible on rerun
Ψ -v2 design (preregistration)	psi-v2-design.md	—
Evidence ledger (chronology)	docs/retrieval-evidence-ledger.md	—

Appendix C. Correction and retraction record

The full chronology, with each correction in the order it occurred, is preserved in docs/retrieval-evidence-ledger.md. The load-bearing corrections, summarised:

1. **Spectral admissibility retracted** (2026-07-30, three passes): first the topical-conduction prediction failed; then the repair was shown to turn the meter green without improving retrieval; then a second agent showed the operator itself was wrong (clique expansion of a hyperedge) and the criterion inverted. §5.2.
2. **An infrastructure failure diagnosed as an architectural one** (2026-07-30): a zero-recall run attributed to fragmentation was :recall-reason :timeout. §5.5(i).
3. **Text-index cause misattributed** (2026-07-30): zero hits blamed on atomic indexing of hyphenated compounds; the real causes are bounded-result starvation and body-only indexing. Correction supplied by the implementing agent. §5.5(iii).
4. **:inclusion-reasons characterised as empty** (Ψ -v2 design note, 2026-07-28): it is populated with a single constant string across the corpus. Same consequence for credit assignment, different repair. §5.5(ii).
5. **:witness-status disclosed as underivable from evidence** (2026-07-30) and excluded as an evaluation signal. Strengthened the same day by audit: the field is not merely underivable but *uncorrelated* with witness records, landing one on each side of the only two cases where such records exist. §2.3, §5.6.
6. **The warrant audit corrected itself twice before reporting** (2026-07-30): two successive brace-walking implementations returned a clean 53-of-53, which was an artifact of never reaching the memory body. Caught by hand-checking the one memory known to carry a witness record. §5.6.

Draft 1 closed 2026-07-30. The living increment log of the pre-draft period is retained in the evidence ledger and is not reproduced here.